

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: III

Name of the Course: Fundamental of Cloud Computing and Bigdata

Course Code: TCS 351

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Discuss the advantages and disadvantages of adopting Cloud Computing in modern IT ecosystems. Illustrate the five <i>essential</i> characteristics of Cloud Computing.	CO1
(b)	Compare and contrast Cloud Computing, Cluster Computing, and Grid Computing with respect to their architecture, scalability, control, resource sharing, and cost-effectiveness. Support your comparison with suitable diagrams	CO1
(c)	Discuss Cloud Computing as defined by NIST Explain all the actors involved in the NIST Cloud Computing Reference Model. Support your Answer with typical architecture	CO1
Q2	(20Marks)	
(a)	Compare and contrast Public, Private, Hybrid, and Community Cloud deployment models with examples of suitable use cases. Support your answer with typical diagram.	CO2
(b)	Illustrate how the cloud architecture differs from traditional client-server computing.	CO2
(c)	Analyze how network architecture, web technologies, and user interfaces influence the scalability and performance of cloud applications.	CO2
Q3	(20Marks)	CO3
(a)	Define and distinguish between IaaS, PaaS, and SaaS, explaining the responsibilities of providers and consumers in each model. Support your Answer with typical block diagram.	CO3
(b)	Discuss Virtualization. Compare and contrast Type-I and Type-II hypervisor with typical block diagram.	CO3
(c)	TechServe Ltd. wants to host internal workloads on a private cloud using open-source tools. Compare Eucalyptus and OpenStack as potential IaaS solutions in terms of architecture, scalability, and management features.	
Q4	(20Marks)	
(a)	Explain the architecture and workflow of Big Data technologies such as Hadoop and Spark, and their role in cloud platforms with proper diagrams	CO5
(b)	Write short notes on Association Rule learning and machine learning with proper Diagrams.	CO5
(c)	Elaborate about the characteristics the technologies used and advantages and disadvantages of Big Data	CO6
Q5	(20Marks)	

(a)	Analyze the importance of Service Level Agreements (SLAs) in cloud computing and discuss their impact on service reliability and customer trust. Discuss the SLA lifecycle with a typical diagram	CO5
(b)	Elaborate the following (any two) 1. AWS S3 2. AWS Lambda 3. AWS RDS	CO6
(c)	A mid-size financial firm is evaluating cloud adoption to reduce hardware costs but is concerned about security, compliance, and downtime. <i>Perform</i> a SWOT analysis and evaluate whether the cloud provides a net economic advantage for the firm.	CO6